

Лабораторная работа №09

Талебу Тенке Франк Устон

2025

Лабораторная работа №09

Лабораторная работа №09

Цель работы

Приобретение практических навыков администрирования сетевых подсистем

На виртуальной машине server войдём под нашим пользователем и откроем терминал. Перейдём в режим суперпользователя и установим необходимые для работы пакеты (рис. @fig-1):

```
[frank@server ~]$ sudo -i
[sudo] password for frank:
[root@server ~]# dnf -y install dovecot telnet
Extra Packages for Enterprise Linux 9 - x86_64 2.3 kB/s | 38 kB 00:16
Extra Packages for Enterprise Linux 9 - x86_64 430 kB/s | 20 MB 00:47
Rocky Linux 9 - BaseOS 392 B/s | 4.1 kB 00:10
Rocky Linux 9 - BaseOS 77 kB/s | 2.5 MB 00:33
Rocky Linux 9 - AppStream 191 B/s | 4.5 kB 00:24
Rocky Linux 9 - AppStream 227 kB/s | 9.5 MB 00:42
Rocky Linux 9 - Extras 270 B/s | 2.9 kB 00:11
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
dovecot x86_64 1:2.3.16-15.el9 appstream 4.7 M
telnet x86_64 1:0.17-85.el9 appstream 63 k
Installing dependencies:
clucene-core x86_64 2.3.3.4-42.20130812.e8e3d20git.el9 appstream 585 k
libexttextcat x86_64 3.4.5-11.el9 appstream 209 k
Transaction Summary
=====
Install 4 Packages

Total download size: 5.6 M
Installed size: 20 M
Downloading Packages:
(1/4): libexttextcat-3.4.5-11.el9.x86_64.rpm 40 kB/s | 209 kB 00:05
(2/4): clucene-core-2.3.3.4-42.20130812.e8e3d20 109 kB/s | 585 kB 00:05
(3/4): telnet-0.17-85.el9.x86_64.rpm 12 kB/s | 63 kB 00:05
(4/4): dovecot-2.3.16-15.el9.x86_64.rpm 3.2 MB/s | 4.7 MB 00:01
```

Рис. 1: Этап 1

```
Downloading Packages:
(1/4): libexttextcat-3.4.5-11.el9.x86_64.rpm      40 kB/s | 209 kB      00:05
(2/4): clucene-core-2.3.3.4-42.20130812.e8e3d20 109 kB/s | 585 kB      00:05
(3/4): telnet-0.17-85.el9.x86_64.rpm             12 kB/s | 63 kB       00:05
(4/4): dovecot-2.3.16-15.el9.x86_64.rpm          3.2 MB/s | 4.7 MB     00:01
-----
Total                                          452 kB/s | 5.6 MB     00:12
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64 1/4
  Installing     : libexttextcat-3.4.5-11.el9.x86_64                2/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64                 3/4
  Installing     : dovecot-1:2.3.16-15.el9.x86_64                 3/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64                 3/4
  Installing     : telnet-1:0.17-85.el9.x86_64                     4/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64                 4/4
  Running scriptlet: telnet-1:0.17-85.el9.x86_64                   4/4
  Verifying      : telnet-1:0.17-85.el9.x86_64                     1/4
  Verifying      : libexttextcat-3.4.5-11.el9.x86_64                2/4
  Verifying      : clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64 3/4
  Verifying      : dovecot-1:2.3.16-15.el9.x86_64                  4/4

Installed:
  clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64      dovecot-1:2.3.16-15.el9.x86_64
  libexttextcat-3.4.5-11.el9.x86_64                          telnet-1:0.17-85.el9.x86_64

Complete!
[root@server ~]#
```

Рис. 2: Этап 2

В Postfix зададим каталог для доставки почты. Сконфигурируем межсетевой экран, разрешив работать службам протоколов POP3 и IMAP (рис. @fig-2):

Восстановим контекст безопасности в SELinux. Перезапустим Postfix и запустим Dovecot (рис. @fig-3):

```
# protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }

# Default values are shown for each setting, it's not required to uncomment
# those. These are exceptions to this though: No sections (e.g. namespace {})
# or plugin settings are added by default, they're listed only as examples.
# Paths are also just examples with the real defaults being based on configure
# options. The paths listed here are for configure --prefix=/usr
# --sysconfdir=/etc --localstatedir=/var

# Protocols we want to be serving.
protocols = imap pop3

# A comma separated list of IPs or hosts where to listen in for connections.
# "*" listens in all IPv4 interfaces, "::" listens in all IPv6 interfaces.
# If you want to specify non-default ports or anything more complex,
# edit conf.d/master.conf.
#listen = *, ::

"/etc/dovecot/dovecot.conf" 101L, 4316B                               24,21  To
```

Рис. 3: Этап 3

На терминале сервера посмотрим имеющуюся почту и mailbox пользователя (рис. @fig-4):

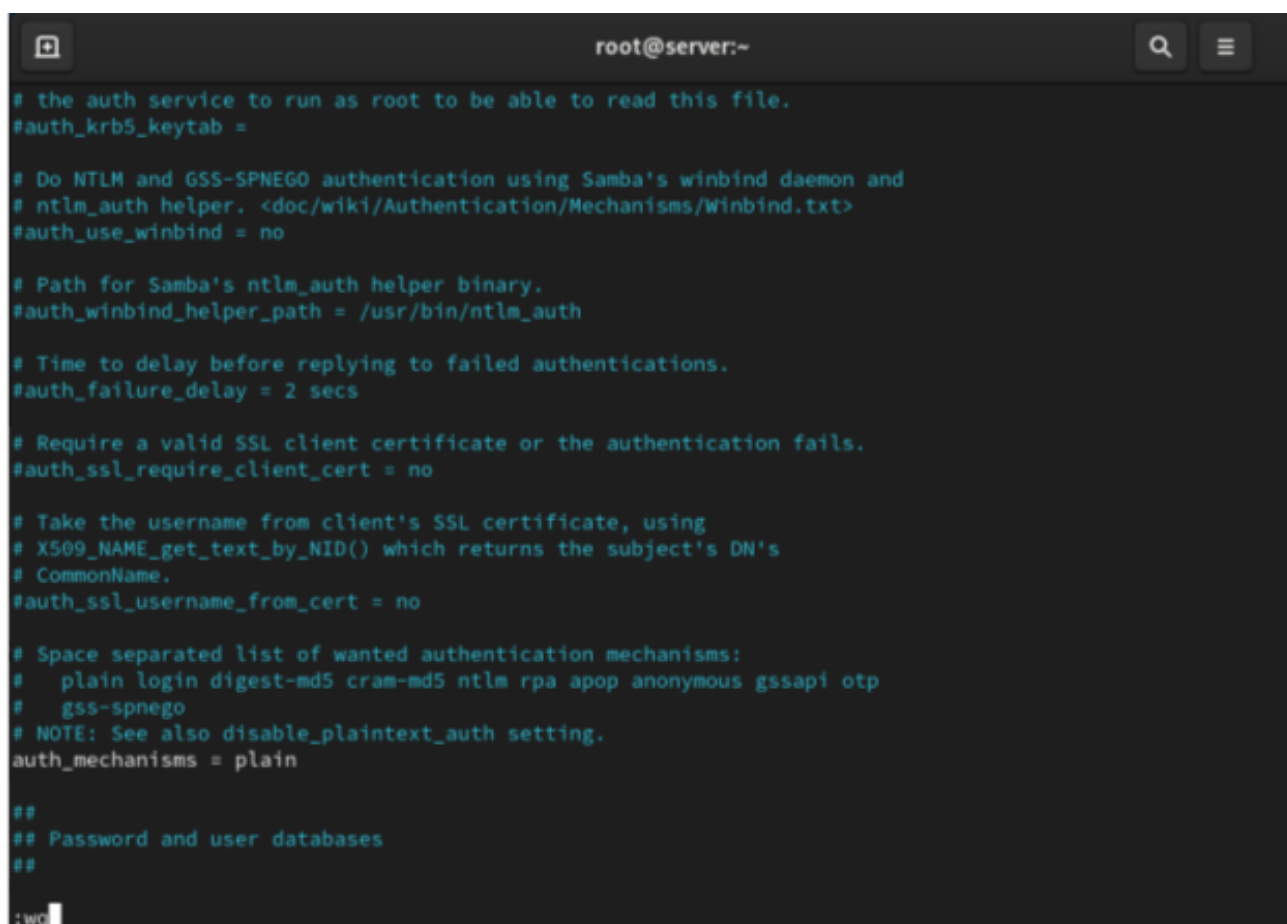
A terminal window with a dark background and light green text. The title bar shows 'root@server:~'. The text inside is a configuration file for Dovecot's authentication, specifically for Samba. It includes comments in Russian and English, and several configuration parameters. The parameters are: #auth_krb5_keytab =, #auth_use_winbind = no, #auth_winbind_helper_path = /usr/bin/ntlm_auth, #auth_failure_delay = 2 secs, #auth_ssl_require_client_cert = no, #auth_ssl_username_from_cert = no, #auth_mechanisms = plain, and ## Password and user databases. The cursor is at the end of the line ':wq'.

Рис. 4: Этап 4

На виртуальной машине client установим почтовый клиент Evolution, запустим и настроим его для работы с нашим почтовым сервером (рис. @fig-5):

Из почтового клиента отправим себе несколько тестовых писем и убедимся, что они доставлены. Параллельно посмотрим сообщения при мониторинге почтовой службы на сервере (рис. @fig-6):

Проверим работу почтовой службы, используя на сервере протокол Telnet: подключимся к почтовому серверу по протоколу POP3, получим список писем, прочитаем первое письмо и удалим второе (рис. @fig-7):

На виртуальной машине server перейдем в каталог для внесения изменений в настройки внутреннего окружения, поместим конфигурационные файлы Dovecot и заменим конфигурационный файл Postfix. Внесем изменения в файл mail.sh на сервере и клиенте (рис. @fig-8):

Этап выполнения работы №9

```
##
# Location for users' mailboxes. The default is empty, which means that Dovecot
# tries to find the mailboxes automatically. This won't work if the user
# doesn't yet have any mail, so you should explicitly tell Dovecot the full
# location.
#
# If you're using mbox, giving a path to the INBOX file (eg. /var/mail/%u)
# isn't enough. You'll also need to tell Dovecot where the other mailboxes are
# kept. This is called the "root mail directory", and it must be the first
# path given in the mail_location setting.
#
# There are a few special variables you can use, eg.:
#
# %u - username
# %n - user part in user@domain, same as %u if there's no domain
# %d - domain part in user@domain, empty if there's no domain
# %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
# mail_location = maildir:~/Maildir
# mail_location = mbox:~/mail:INBOX=/var/mail/%u
# mail_location = mbox:/var/mail/%d/%n:INDEX=/var/indexes/%d/%n/%n
#
# <doc/wiki/MailLocation.txt>
#
mail_location = maildir:~/Maildir

# If you need to set multiple mailbox locations or want to change default
# namespace settings, you can do it by defining namespace sections.
#
# You can have private, shared and public namespaces. Private namespaces
# are for user's personal mails. Shared namespaces are for accessing other
# users' mailboxes that have been shared. Public namespaces are for shared
# mailboxes that are managed by sysadmin. If you create any shared or public
# namespaces you'll typically want to enable ACL plugin also, otherwise all
# users can access all the shared mailboxes, assuming they have permissions
# on filesystem level to do so.
namespace inbox {
```

Рис. 5: Этап 5

```
[root@server ~]#
[root@server ~]# postconf -e 'home_mailbox = Maildir/'
[root@server ~]#
[root@server ~]#
```

Рис. 6: Этап 6

```

[root@server ~]#
[root@server ~]# postconf -e 'home_mailbox = Maildir/'
[root@server ~]#
[root@server ~]#
[root@server ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp2
bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testn
et bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd cond
or-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls dock
er-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy fr
eeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master
git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs is
csi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver
kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure kube-nodep
ort-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap lda
ps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcac
he minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboar
d nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pncd pmpoxy
pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3ne
tsrv ptp pulseaudio puppetmaster quassel rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt
-master samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snm
ptrap spideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn syncthing syncthing-gui syncthin
g-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client v
dsm vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-
discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerot
ier
[root@server ~]# firewall-cmd --add-service=pop3 --permanent
success
[root@server ~]# firewall-cmd --add-service=pop3s --permanent
success
[root@server ~]# firewall-cmd --add-service=imap --permanen
success
[root@server ~]# irewall-cmd --add-service=imaps --permanent
bash: irewall-cmd: command not found...
^[[A^[[B^[[B
^C
[root@server ~]# firewall-cmd --add-service=imaps --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# firewall-cmd --list-services
cockpit dhcpv6-client imap imaps pop3 pop3s smtp ssh
[root@server ~]#

```

Рис. 7: Этап 7

```

[root@server ~]#
[root@server ~]# restorecon -vR /etc
[root@server ~]# systemctl restart postfix
[root@server ~]# systemctl enable dovecot
Created symlink /etc/systemd/system/multi-user.target.wants/dovecot.service → /usr/lib/systemd/system/dovecot.
service.
[root@server ~]# systemctl start dovecot
[root@server ~]#

```

Рис. 8: Этап 8

```

[root@server ~]# MAIL=~/.Maildir mai
bash: mai: command not found...
[root@server ~]# MAIL=~/.Maildir mail
s-nail: No mail for root at /root/.Maildir
s-nail: /root/.Maildir: No such entry, file or directory
[root@server ~]#
[root@server ~]# doveadm mailbox list -u user
INBOX
[root@server ~]# doveadm mailbox list -u frank
INBOX
[root@server ~]#

```

Рис. 9: Этап 9

Welcome
Restore from Backup
Identity
Receiving Email
Sending Email
Account Summary
Done

Please enter your name and email address below. The "optional" fields below do not need to be filled in, unless you wish to include this information in email you send.

Required Information

Full Name:

Email Address:

Optional Information

Reply-To:

Organization:

Aliases:

Welcome
Restore from Backup
Identity
Receiving Email
Receiving Options
Sending Email
Account Summary
Done

Receiving Email

Server Type:

Description: For reading and storing mail on IMAP servers.

Configuration

Server: Port:

Username:

Security

Encryption method:

Authentication

Рис. 10: Этап 10

Этап выполнения работы №10

Этап выполнения работы №11

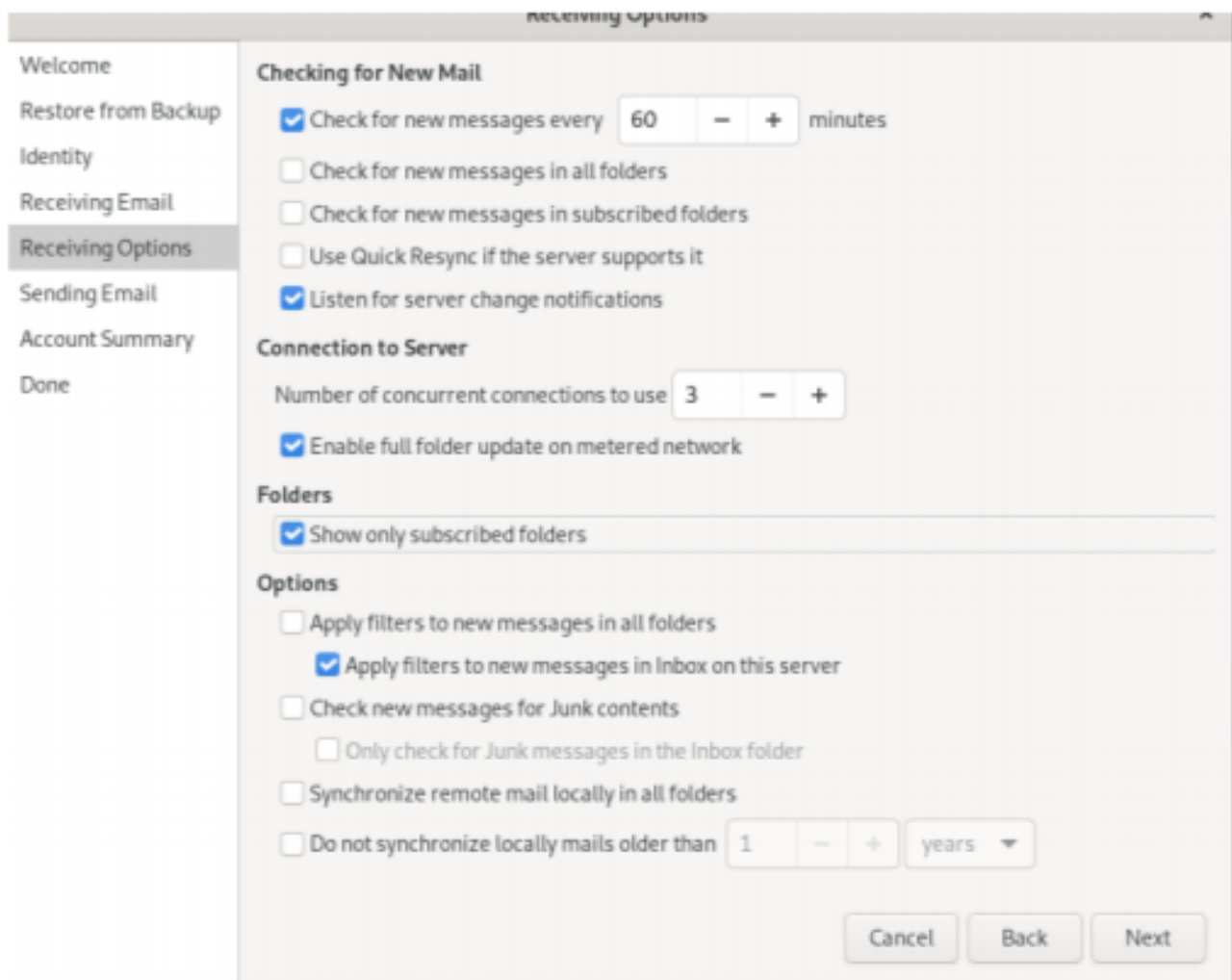


Рис. 11: Этап 11

Этап выполнения работы №12

Этап выполнения работы №13

Этап выполнения работы №14

Этап выполнения работы №15

Этап выполнения работы №16

Welcome

Restore from Backup

Identity

Receiving Email

Receiving Options

Sending Email

Account Summary

Done

Sending Email

Server Type:

SMTP

Description:

For delivering mail by connecting to a remote mailhub using SMTP.

Configuration

Server:

mail.frank.net

Port:

25

☐

Server requires authentication

Security

Encryption method:

TLS on a dedicated port

Authentication

Type:

Check for Supported TypesPLAIN

Username:

frank

Account Summary

×

Welcome

Restore from Backup

Identity

Receiving Email

Receiving Options

Sending Email

Account Summary

Done

This is a summary of the settings which will be used to access your mail.

Account Information

Name:

The above name will be used to identify this account.
Use for example, "Work" or "Personal".

Personal Details

Full Name: frank

Email Address: frank@frank.net

	Receiving	Sending
Server Type:	imapx	smtp
Server:	mail.frank.net	mail.frank.net
Username:	frank	frank
Security:	STARTTLS	TLS

Рис. 12: Этап 12

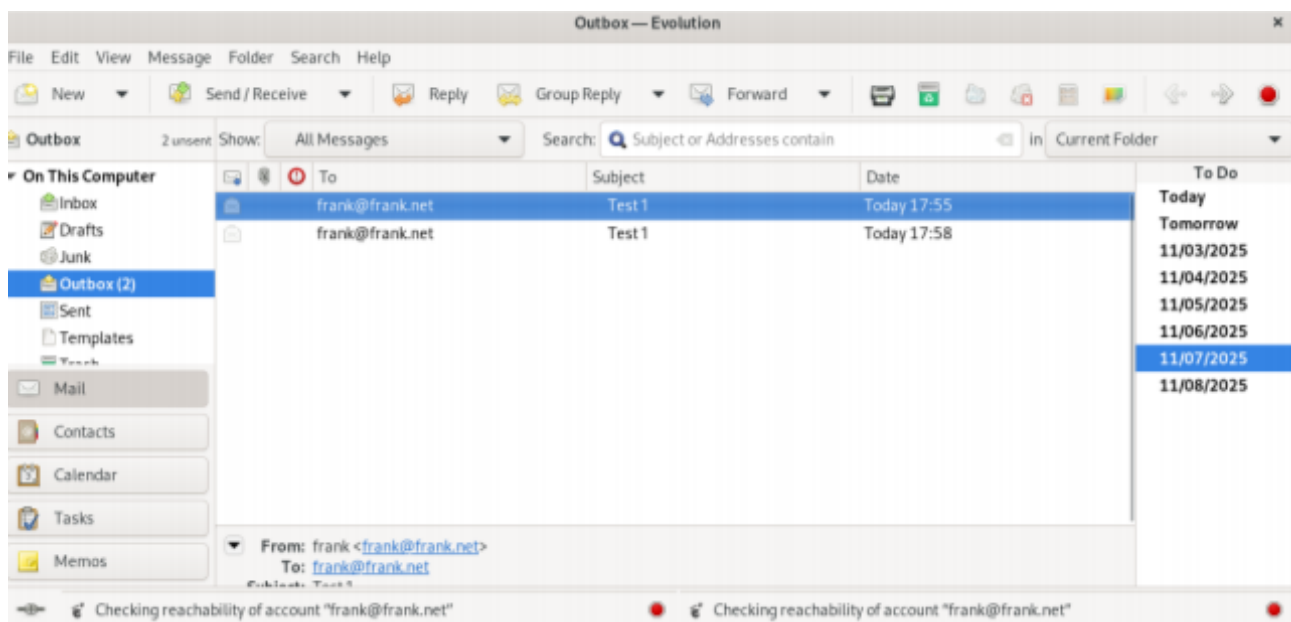


Рис. 13: Этап 13

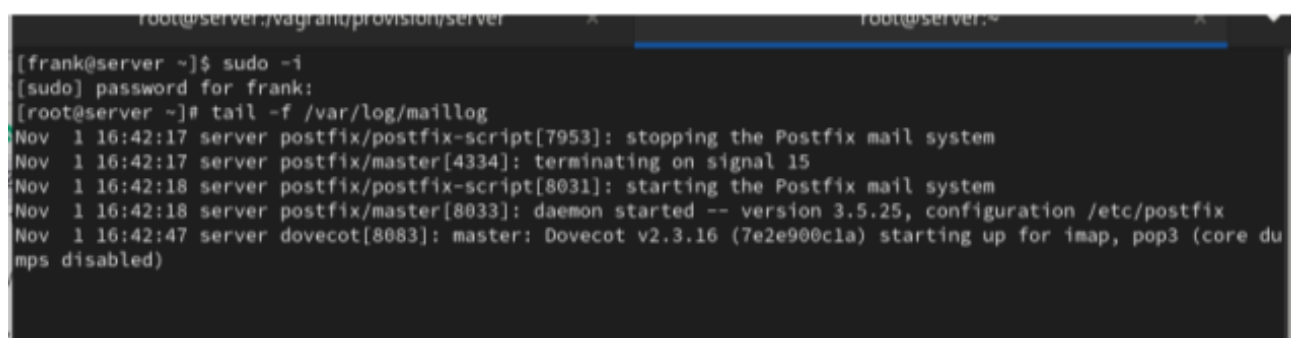


Рис. 14: Этап 14

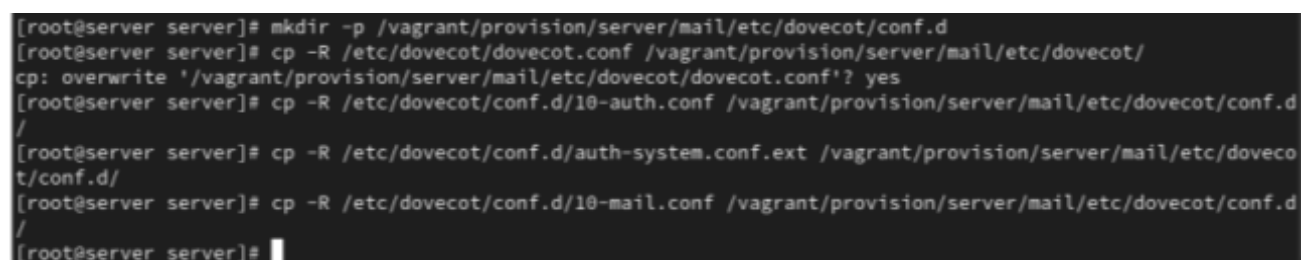


Рис. 15: Этап 15

```

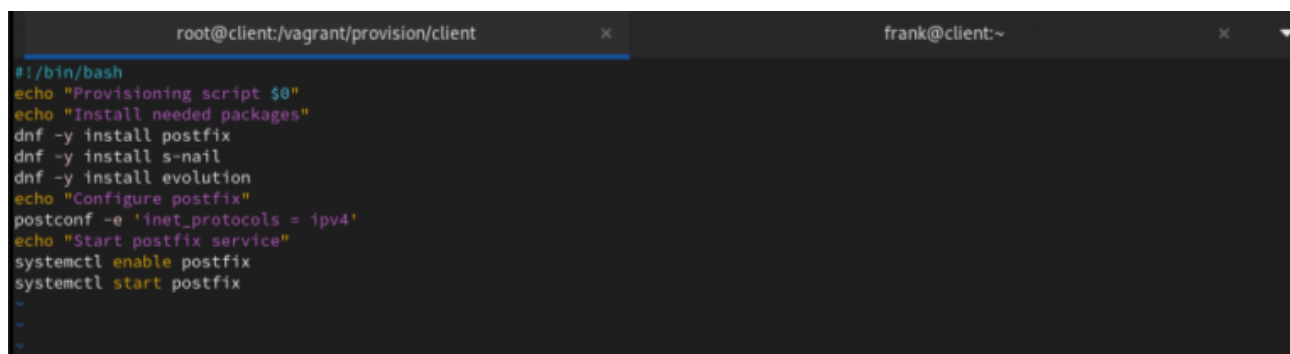
#!/bin/bash
cho "Provisioning script $0"
cho "Install needed packages"
nf -y install postfix
nf -y install s-nail
nf -y install Dovecot Telnet
cho "Copy configuration files"
cp -R /vagrant/provision/server/mail/etc/* /etc
cho "Configure firewall"
firewall-cmd --add-service=smtp --permanent
firewall-cmd --reload
estorecon -vR /etc
firewall-cmd --add-service=pop3 --permanent
firewall-cmd --add-service=pop3s --permanent
firewall-cmd --add-service=imap --permanent
firewall-cmd --add-service=imaps --permanent
firewall-cmd --reload
firewall-cmd --list-services

cho "Start postfix service"
systemctl enable postfix
systemctl start postfix
systemctl restart postfix
systemctl enable dovecot
systemctl start dovecot
cho "Configure postfix"
ostconf -e 'mydomain = user.net'
ostconf -e 'myorigin = $mydomain'
ostconf -e 'inet_protocols = ipv4'
ostconf -e 'inet_interfaces = all'
ostconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost,

```

Рис. 16: Этап 16

Этап выполнения работы №17



```

root@client:/vagrant/provision/client  x  frank@client:~
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
dnf -y install evolution
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
>
>
>

```

Рис. 17: Этап 17

Выводы

В ходе выполнения лабораторной работы были приобретены практические навыки

Спасибо за внимание!

Вопросы?